

from google.colab import files
uploaded = files.upload()

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Saving youtube analysis.csv to youtube analysis.csv

CODE 1

```
import pandas as pd

# Load CSV file
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

# Keep only needed columns
df = df[["Youtuber", "subscribers"]]

# Get top 10 channels by subscribers
top10 = df.nlargest(10, "subscribers")

# Display result
print("Top 10 YouTube Channels by Subscribers:")
print(top10)
```

Top 10 YouTube Channels by Subscribers:

	Youtuber	subscribers
0	T-Series	245000000.0
1	YouTube Movies	170000000.0
2	MrBeast	166000000.0
3	Cocomelon - Nursery Rhymes	162000000.0
4	SET India	159000000.0
5	Music	119000000.0
6	ýýý Kids Diana Show	112000000.0
7	PewDiePie	111000000.0
8	Like Nastya	106000000.0
9	Vlad and Niki	98900000.0

CODE 2

```
import pandas as pd
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

# Keep needed columns
df = df[["category", "subscribers"]]

# Remove missing values
#df = df.dropna()

# Convert subscribers to numeric
#df["subscribers"] = pd.to_numeric(df["subscribers"], errors="coerce")

# Calculate average subscribers per category
avg_subs = df.groupby("category")["subscribers"].mean()

# Find category with highest average
top_category = avg_subs.idxmax()
top_value = avg_subs.max()
print("Category with highest average subscribers:", top_category)

print("Average subscribers:", int(top_value))
```

Category with highest average subscribers: Shows
Average subscribers: 41615384

CODE 3

```
import pandas as pd

# Load data
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

# Average uploads per category
avg_uploads = df.groupby("category")["uploads"].mean()

print(avg_uploads)
```

category	
Autos & Vehicles	1550.666667
Comedy	1202.557143
Education	3087.086957
Entertainment	12052.445378
Film & Animation	2861.844444
Gaming	4285.273684
Howto & Style	1695.500000
Movies	3553.000000
Music	2325.945813
News & Politics	112484.384615
Nonprofits & Activism	102912.000000
People & Blogs	9256.793893
Pets & Animals	3562.800000
Science & Technology	2114.058824
Shows	27443.692308
Sports	19129.833333
Trailers	6839.000000
Travel & Events	766.000000

Name: uploads, dtype: float64

CODE 4

```
import pandas as pd

# Load dataset
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

df = df[["Country"]]

country_counts = df["Country"].value_counts()

# Top 5 countries
top5_countries = country_counts.head(5)

print("Top 5 countries with highest number of YouTube channels:")
print(top5_countries)
```

Top 5 countries with highest number of YouTube channels:

Country	
United States	315
India	169
Brazil	62
United Kingdom	44
Mexico	33

Name: count, dtype: int64

CODE 5

```
import pandas as pd

# Load CSV file
df = pd.read_csv("youtube analysis.csv", encoding="latin-1" )

# Distribution of channel types across categories
distribution = pd.crosstab(df['category'], df['channel_type'])

print(distribution)
```

channel_type	Animals	Autos	Comedy	Education	Entertainment	Film \	
category							
Autos & Vehicles	0	2	0	0	1	0	
Comedy	0	0	39	0	20	1	
Education	0	0	0	36	3	2	
Entertainment	0	1	5	2	168	6	
Film & Animation	0	0	1	2	16	17	
Gaming	0	1	1	0	18	2	
Howto & Style	0	0	0	0	7	0	
Movies	0	0	0	0	0	2	
Music	0	0	0	1	12	5	
News & Politics	0	0	0	0	3	0	
Nonprofits & Activism	0	0	0	0	0	0	
People & Blogs	1	0	5	3	26	3	
Pets & Animals	2	0	0	0	1	0	
Science & Technology	0	0	0	0	4	0	
Shows	0	0	1	2	6	2	
Sports	0	0	0	0	1	0	
Trailers	0	0	0	0	1	0	
Travel & Events	0	0	0	0	1	0	

channel_type	Games	Howto	Music	News	Nonprofit	People	Sports	\
category								
Autos & Vehicles	0	0	0	0	0	0	0	

Comedy	3	0	0	0	0	5	0
Education	2	0	0	0	0	3	0
Entertainment	11	0	22	3	0	11	0
Film & Animation	3	0	3	0	0	2	0
Gaming	64	0	0	0	0	6	0
Howto & Style	0	27	0	0	0	2	0
Movies	0	0	0	0	0	0	0
Music	2	0	175	1	0	2	1
News & Politics	0	0	1	22	0	0	0
Nonprofits & Activism	0	0	0	0	2	0	0
People & Blogs	8	9	8	4	0	58	1
Pets & Animals	0	0	0	0	0	0	0
Science & Technology	0	0	0	0	0	0	0
Shows	0	0	1	0	0	0	0
Sports	0	0	0	0	0	0	11
Trailers	0	0	1	0	0	0	0
Travel & Events	0	0	0	0	0	0	0

channel_type	Tech
category	
Autos & Vehicles	0
Comedy	0
Education	0
Entertainment	1
Film & Animation	0
Gaming	1
Howto & Style	1
Movies	0
Music	0
News & Politics	0
Nonprofits & Activism	0
People & Blogs	1
Pets & Animals	0

CODE 6

<pre>import pandas as pd # Load dataset df = pd.read_csv("youtube analysis.csv" , encoding="latin-1") # Correlation between subscribers and video views correlation = df['subscribers'].corr(df['video views']) print("Correlation:", correlation)</pre>
Correlation: 0.7481786016237689

CODE 7

```
import pandas as pd

# Load dataset
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

# Average monthly earnings per category
#earnings_by_category = df.groupby('category')['monthly earnings'].mean()
earnings_by_category = df.groupby('category')[['highest_monthly_earnings', 'lowest_monthly_earnings']].mean()
print(earnings_by_category)
```

	highest_monthly_earnings	lowest_monthly_earnings
category		
Autos & Vehicles	1.1909900e+06	74966.666667
Comedy	6.597635e+05	41332.285714
Education	7.518043e+05	46863.239348
Entertainment	6.293549e+05	39393.724370
Film & Animation	7.489841e+05	46802.533556
Gaming	2.720201e+05	16965.790316
Howto & Style	1.920441e+05	12022.350500
Movies	4.547000e+05	28400.000000
Music	5.535735e+05	34671.696798
News & Politics	6.426320e+05	40192.625000
Nonprofits & Activism	3.904000e+05	24400.000000
People & Blogs	5.357493e+05	33485.993969
Pets & Animals	6.355516e+05	39980.800000
Science & Technology	2.020432e+05	12635.411765
Shows	2.037662e+06	126961.538462
Sports	9.813583e+05	60783.333333
Trailers	3.619000e+05	22600.000000
Travel & Events	1.240000e+05	7800.000000

CODE 8

```
import pandas as pd
import matplotlib.pyplot as plt

# Load CSV file
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

# Histogram across all channels
df['subscribers_for_last_30_days'].hist()

plt.xlabel("Subscribers Gained in Last 30 Days")
plt.ylabel("Number of Channels")
plt.title("Overall Trend of Subscribers Gained in Last 30 Days Across All Channels")

plt.show()
```

Overall Trend of Subscribers Gained in Last 30 Days Across All Channels

Subscribers Gained (x1e6)	Number of Channels
0 - 1	600
1 - 2	50
2 - 3	10
3 - 4	5
4 - 5	2
5 - 6	1
6 - 7	1
7 - 8	1

CODE 9

```
import pandas as pd

# Load dataset
df = pd.read_csv("youtube_analysis.csv", encoding="latin-1")

# Clean currency columns
df["lowest_yearly_earnings"] = df["lowest_yearly_earnings"].replace(r'[\$,]', '', regex=True).astype(float)
df["highest_yearly_earnings"] = df["highest_yearly_earnings"].replace(r'[\$,]', '', regex=True).astype(float)

# Function to detect outliers using IQR
def find_outliers(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    return (column < lower) | (column > upper)

# Detect outliers
lowest_outliers = df[find_outliers(df["lowest_yearly_earnings"])]
highest_outliers = df[find_outliers(df["highest_yearly_earnings"])]

print("Outliers in Lowest Yearly Earnings:\n")
print(lowest_outliers[["Youtuber", "lowest_yearly_earnings"]])

print("\nOutliers in Highest Yearly Earnings:\n")
print(highest_outliers[["Youtuber", "highest_yearly_earnings"]])
```

```
Outliers in Lowest Yearly Earnings:
```

	Youtuber	lowest_yearly_earnings
0	T-Series	6800000.0
2	MrBeast	4000000.0
3	Cocomelon - Nursery Rhymes	5900000.0
4	SET India	5500000.0
6	ýýý Kids Diana Show	2200000.0
...	...	...

```
970         _vector_      270000.0
990         Natan por Aiz 170000.0
995         WWE           210000.0
999         Infobells - Hindi 240000.0
1002        Supercar Blondie 110000.0

[105 rows x 2 columns]

Outliers in Highest Yearly Earnings:

      Youtuber  highest_yearly_earnings
0      T-Series      108400000.0
2      MrBeast      64700000.0
3  Cocomelon - Nursery Rhymes      94800000.0
4      SET India      87500000.0
6      yyy Kids Diana Show      35100000.0
..         ...
908      Susy Mouriz      17700000.0
970         _vector_      43400000.0
990         Natan por Aiz      26500000.0
995         WWE           34300000.0
999         Infobells - Hindi      39200000.0

[101 rows x 2 columns]
```

CODE 10

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

# Count channels created each year
yearly_distribution = df["created_year"].value_counts().sort_index()

print("Distribution of channel creation by year:\n")
print(yearly_distribution)

# Plot trend
plt.figure()
yearly_distribution.plot(kind="line")
plt.xlabel("Year")
plt.ylabel("Number of Channels Created")
plt.title("Trend of YouTube Channel Creation Over Time")
plt.show()
```

Distribution of channel creation by year:

```
created_year
1970.0      1
2005.0     24
2006.0     93
2007.0     52
2008.0     46
2009.0     52
2010.0     48
2011.0     82
2012.0     68
2013.0     78
2014.0    100
2015.0     73
2016.0     78
2017.0     68
2018.0     46
2019.0     34
2020.0     30
2021.0     23
2022.0      5
Name: count, dtype: int64
```

CODE 11

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

# number of channels per country (count Youtuber)
channels = df.groupby("Country")["Youtuber"].count()

# education per country
education = df.groupby("Country")["Gross tertiary education enrollment (%)"].first()

# combine
data = pd.concat([channels, education], axis=1).dropna()
data.columns = ["num_channels", "education"]

# convert % to numeric
data["education"] = data["education"].astype(str).str.replace("%","").astype(float)

# correlation
print("Correlation:", data["num_channels"].corr(data["education"]))

# scatter plot
plt.figure()
plt.scatter(data["education"], data["num_channels"])
plt.xlabel("Gross tertiary education enrollment (%)")
plt.ylabel("Number of YouTube Channels")
plt.show()
```

Correlation: 0.1121990458668538

CODE 12

```
import pandas as pd
import matplotlib.pyplot as plt

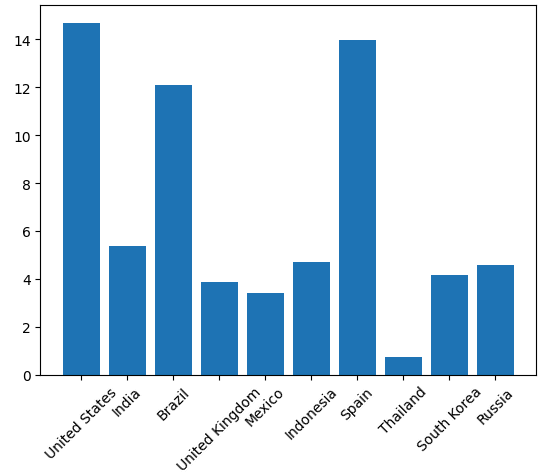
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

top10 = df.groupby("Country")["Youtuber"].count().sort_values(ascending=False).head(10)
unemp = df.groupby("Country")["Unemployment rate"].first()

data = pd.concat([top10, unemp], axis=1).dropna()

plt.bar(data.index, data["Unemployment rate"])
```

```
plt.xticks(rotation=45)
plt.show()
```



CODE 13

```
import pandas as pd

df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

avg_urban = df["Urban_population"].mean()

print("Average urban population (%):", avg_urban)
```

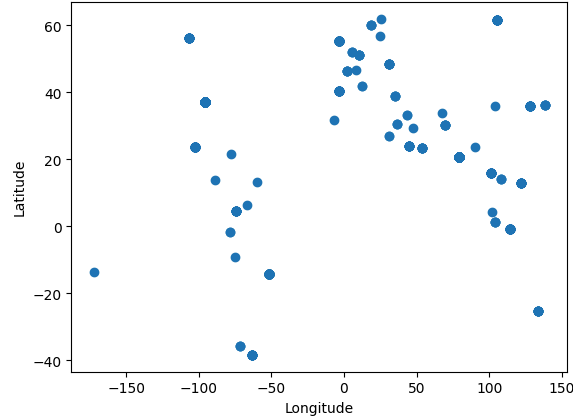
Average urban population (%): 223974718.82045454

CODE 14

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

plt.scatter(df["Longitude"], df["Latitude"])
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.show()
```



CODE 15

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

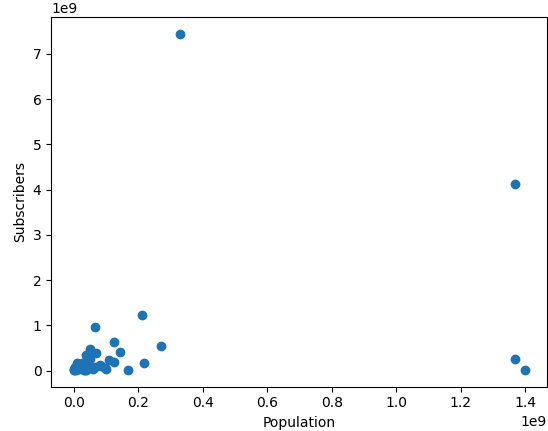
subs = df.groupby("Country")["subscribers"].sum()
pop = df.groupby("Country")["Population"].first()

data = pd.concat([subs, pop], axis=1).dropna()

print("Correlation:", data["subscribers"].corr(data["Population"]))

plt.scatter(data["Population"], data["subscribers"])
plt.xlabel("Population")
plt.ylabel("Subscribers")
plt.show()
```

Correlation: 0.340759532520351



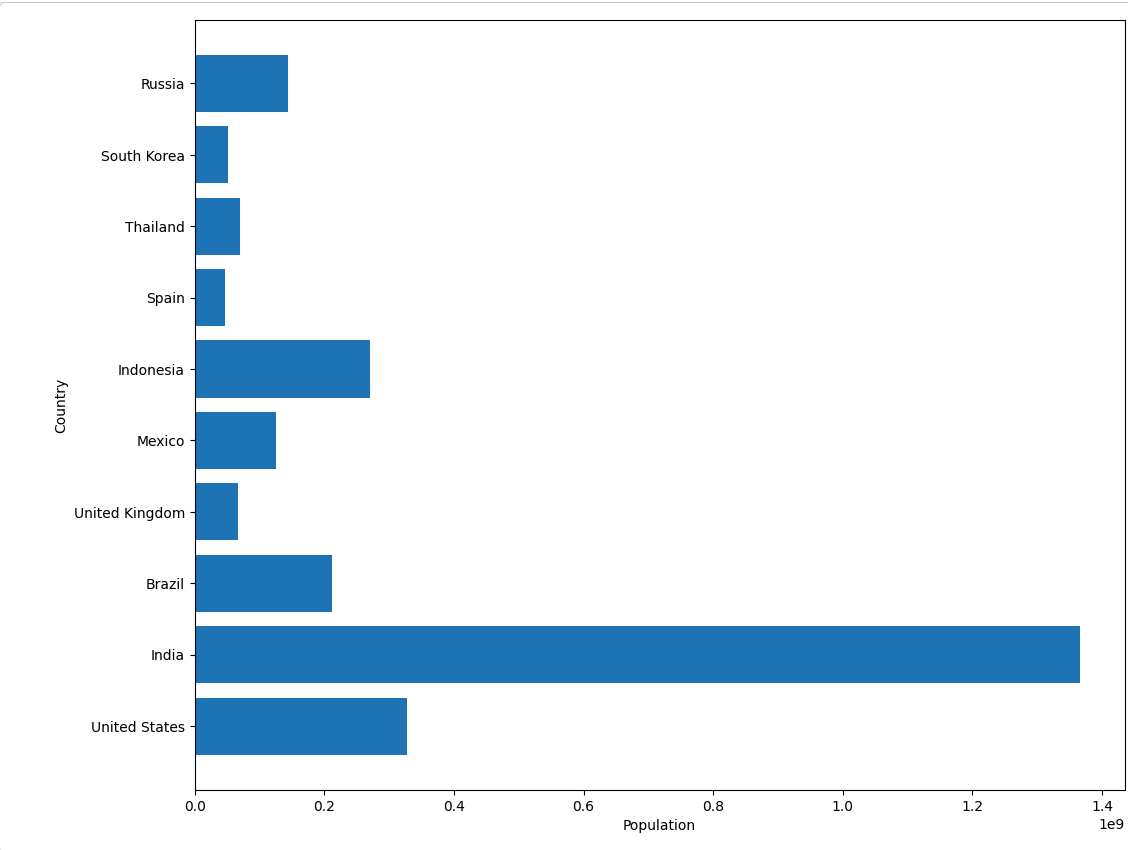
CODE 16

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

top10 = df.groupby("Country")["Youtuber"].count().sort_values(ascending=False).head(10)
pop = df.groupby("Country")["Population"].first().loc[top10.index]

plt.figure(figsize=(12, 10))
plt.barh(top10.index, pop)
plt.xlabel("Population")
plt.ylabel("Country")
plt.show()
```

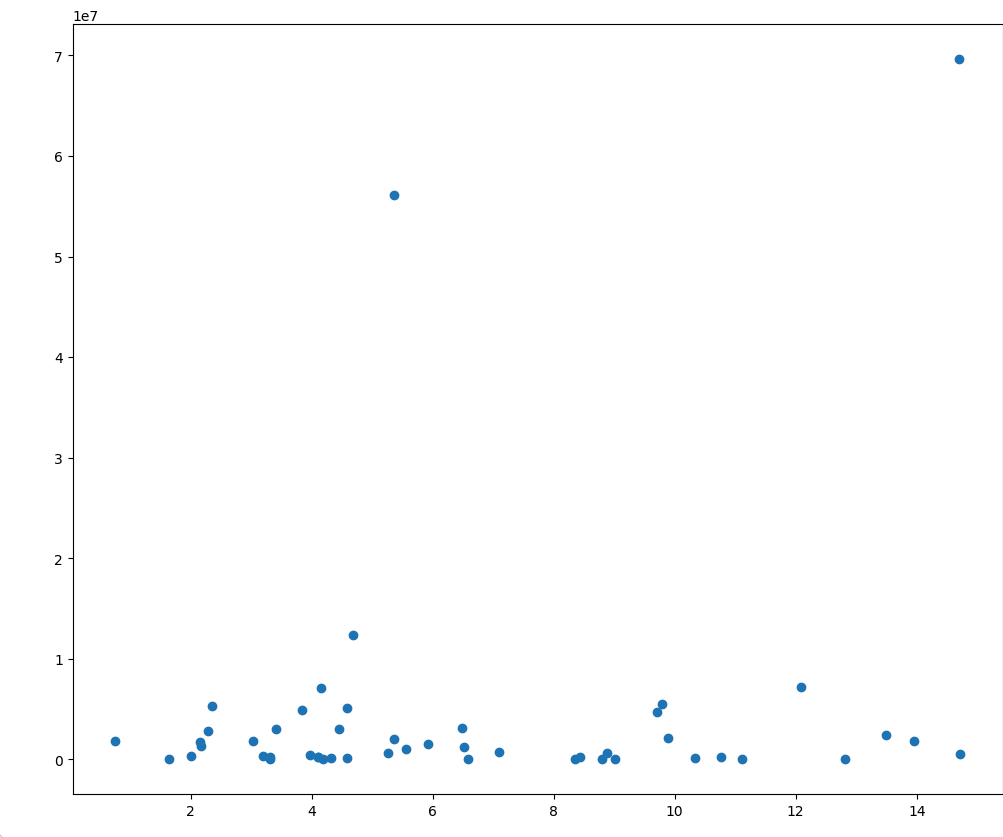


CODE 17

```
import pandas as pd
import matplotlib.pyplot as plt

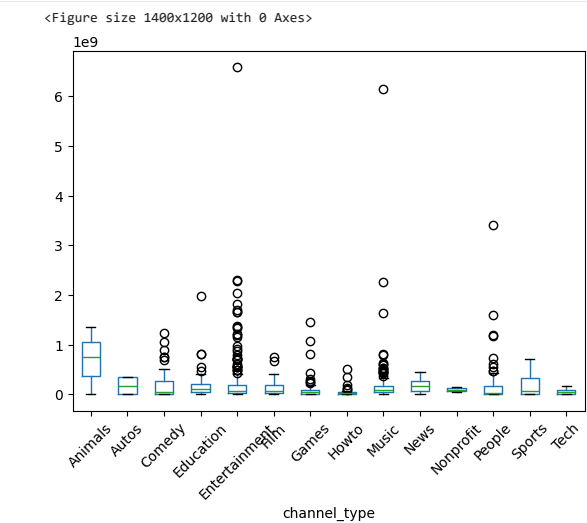
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

data = pd.concat([
    df.groupby("Country")["subscribers_for_last_30_days"].sum(),
    df.groupby("Country")["Unemployment rate"].first()
], axis=1).dropna()
plt.figure(figsize=(12 , 10))
plt.scatter(data["Unemployment rate"], data["subscribers_for_last_30_days"])
plt.show()
```



CODE 18

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")
plt.figure(figsize=(14 , 12))
df.boxplot(column="video_views_for_the_last_30_days", by="channel_type", grid=False)
plt.xticks(rotation=45)
plt.title("")
plt.suptitle("")
plt.show()
```



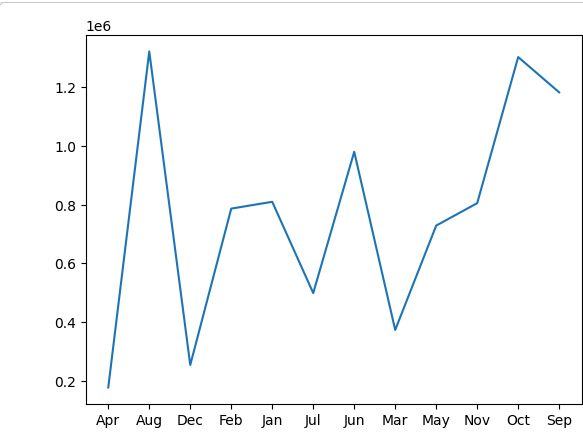
Double-click (or enter) to edit

CODE 19

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

uploads = df.groupby("created_month")["uploads"].sum()

plt.plot(uploads.index, uploads.values)
plt.show()
```



CODE 20

```
import pandas as pd
from datetime import datetime

df = pd.read_csv("youtube analysis.csv", encoding="latin-1")

df["created_year"] = pd.to_numeric(df["created_year"], errors="coerce")
df["created_month"] = pd.to_numeric(df["created_month"], errors="coerce")

now = datetime.now()
months = (now.year - df["created_year"]) * 12 + (now.month - df["created_month"])

avg_subs_per_month = (df["subscribers"] / months.replace(0, pd.NA)).mean()

print(avg_subs_per_month)
```

nan